

what is the solution of $y''(x) = i*y/x$?



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The general solution to the second-order differential equation $y''(x) = \frac{i \cdot y}{x}$ is given by $y(x) = \sqrt{x} \left[C_1 J_1 \left(2\sqrt{-ix} \right) + C_2 Y_1 \left(2\sqrt{-ix} \right) \right]$, where C_1 and C_2 are arbitrary constants.

Step-by-Step Breakdown

1. Rearrange the Equation

Multiply both sides by x^2 to put it into a form related to Bessel's differential equation:

$$x^2 y'' - ixy = 0$$

2. Substitute Variables

To transform this into standard form, we define a new independent variable z and dependent function $u(z)$ using the transformations:

- $z = 2\sqrt{-ix} = \sqrt{2}(1-i)\sqrt{x}$
- $y(x) = \sqrt{x} \cdot u(z)$

3. Convert to Bessel's Equation

Applying these substitutions and computing the derivatives with respect to z yields the **Bessel differential equation of order 1**:

$$z^2 \frac{d^2 u}{dz^2} + z \frac{du}{dz} + (z^2 - 1)u = 0$$

4. Formulate the General Solution

The linearly independent solutions to this equation are the Bessel functions of the first kind $J_1(z)$ and second kind $Y_1(z)$:

$$u(z) = C_1 J_1(z) + C_2 Y_1(z)$$

Substituting z and $u(z)$ back into terms of x gives the final general solution:

$$y(x) = \sqrt{x} \left[C_1 J_1 \left(2\sqrt{-ix} \right) + C_2 Y_1 \left(2\sqrt{-ix} \right) \right]$$